

RANA MUHAMMAD UZAIR

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PROFESSIONAL SUMMARY

Full Stack & AI/ML Engineer with hands-on experience building transformer-based generative AI models, enterprise-grade security platforms, MERN stack web applications, and machine learning pipelines. Trained deep learning models from scratch, architected full-stack systems end-to-end, and deployed live production applications. Currently pursuing B.S. Computer Science at CUNY John Jay College.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java

Frontend: React.js, Next.js, Vite, HTML5, CSS3, Tailwind CSS, Framer Motion

Backend: Node.js, Express.js, FastAPI, Flask, Spring Boot, REST APIs, SSE

AI/ML: PyTorch, TensorFlow/Keras (LSTM), XGBoost, Scikit-learn, DeepFace, Ollama (LLM), Pandas, NumPy, Matplotlib

Tools & Databases: Git, GitHub, MongoDB, SQLite, SQL, Vercel, Netlify

PROJECTS

NexaGuard | FastAPI · React · TensorFlow (LSTM) · XGBoost · DeepFace · Ollama · SQLite

github.com/UZAIR676/NexaGuard

- Built an enterprise-grade AI-powered financial security platform — fraud detection, live stock analytics, ML price prediction, biometric auth, and an AI financial advisor, all in one full-stack app
- Custom LSTM + XGBoost hybrid model trained on 16 engineered features (RSI, MACD, Bollinger Bands, ATR, momentum) — outputs BUY/HOLD/SELL signal with confidence score; scans 15 top stocks in real-time
- Transaction fraud detection with Random Forest: auto-approves (<40%), email confirmation (40–70%), or blocks + alerts (>70%) — includes Geo-IP impossible travel detection via MaxMind GeoLite2
- Face biometric authentication using DeepFace (100% local, no cloud) — liveness detection, embedding storage, multi-frame anti-spoof; triggered as override for high-risk transactions
- AI Financial Advisor powered by Qwen 2.5 7B (Ollama, runs locally) — injects live market data + technical indicators + ML predictions into every response via SSE streaming
- Role-based access control (User / Analyst / Admin), OTP email verification, SHA-256 password hashing, and per-user velocity abuse detection

Mood Music AI | PyTorch · Transformer · Flask · MAESTRO · pretty_midi

github.com/UZAIR676/mood-music-ai

- Built a transformer-based generative AI from scratch that composes original piano music — trained on Google's MAESTRO dataset (200+ hours of professional recordings)
- Custom tokenizer encoding each note as 4 tokens (pitch, velocity, time, duration) across a 364-token vocabulary; 3-layer decoder, 4 attention heads, 128-dim embeddings, 512 FFN
- Mood control via musical scale filtering (Major, Minor, Pentatonic, Diminished) with per-mood temperature sampling; Flask REST API + web UI for real-time generation and MIDI download

TaskPilot | Java · Spring Boot · React · Google Gemini AI

task-pilot-lac.vercel.app

- AI-powered task manager integrating Google Gemini API for natural language task creation, AI summarization, and subtask breakdown — Spring Boot REST API + React.js frontend, live on Vercel

Car Price Prediction Web App | Python · Flask · React · Random Forest · Scikit-learn

github.com/UZAIR676/car_prediction-

- ML web app predicting used car prices with 92% accuracy using Random Forest — Flask REST API + React.js frontend with dark/light mode toggle

E-Commerce Platform | React · Node.js · MongoDB · Tailwind CSS

store-app-ten-rho.vercel.app

- Full-stack e-commerce store with authentication, product management, cart, and payment processing — live on Vercel

WORK EXPERIENCE

Amazon — Blue Badge Employee | New York, NY | 2025 – Present

- Managed high-volume operational workflows while pursuing B.S. Computer Science full-time
- Developed time management, cross-functional teamwork, and workplace discipline in a fast-paced environment

EDUCATION

B.S. Computer Science | LaGuardia Community College – John Jay College, CUNY | GPA: 3.2

A.S. Computer Science | LaGuardia Community College, CUNY